

Absolutely — here's the full change log of what I modified in this session, grouped by file.

routes/batch_routes.py

Fixed the crash in batch_generate:

initialized num_sources

initialized is_force_crossing

initialized road_curve_a

removed duplicate delay assignment

Updated multi-source busy-road generation behavior:

moved away from strict deterministic overlap patterns

iterated slot assignment strategy (overlap allowed, then reverted to separated, then improved)

added/adjusted overtake-style Bézier logic

enforced non-straight behavior (parabola/Bézier only in multi-source branch)

strengthened curvature checks to avoid nearly straight paths

adjusted parabola confinement to use $\text{abs}(\text{speed}) * \text{duration}$ instead of fixed span math

Added B9 intersection benchmark backend branch:

detects intersection_benchmark + B9

generates plus-intersection traffic (E-W + N-S approaches)

calls generate_multi_object_clip with intersection geometry context

made B9 intersection not interleaved away by even-index gating

Intersection-specific logic changes:

car count now follows selected num_sources (not hardcoded 4)

observer position changed multiple times; final logic uses equal x/y distance in Q1:

from intersection_observer_distance (default 2m), position ($d/\sqrt{2}$, $d/\sqrt{2}$)

added lane inner margin and inner offsets so paths stay inside bounds

clamped Bézier control points for intersection lanes

reduced random tilt in intersection to avoid lane escape

added per-approach de-overlap spacing for repeated vehicles

visualization/plot_utils.py

Refactored save_combined_path_plot heavily:

removed decorative asphalt/grass style road background

replaced with minimal road guides (edges + median)

later added intersection-specific plus-road guide drawing (E-W and N-S edges/medians)

Fixed bugs:

fixed `y_med` unbound local error in non-rotated branch

corrected road rotation math/pivot behavior for non-zero `road_angle`

Road geometry rendering updates:

introduced curved road centerline logic with multiple modes (parabola, bezier, etc.)

computed lane edges via normal offsets from centerline

added defaults for non-zero visible curvature

Viewport/scaling iterations:

multiple y-axis and aspect adjustments for readability vs geometric fidelity

final important fix: dedicated square viewport path for intersection mode (instead of strip-road y-cap logic)

Vehicle styling updates:

thinned vehicle lines

later removed start/end dots from combined plots (and single path plots too)

`compute_path_points` updates:

added per-vehicle transform knobs:

`road_curve_blend`

`road_angle_offset`

`global_curve_scale`

added optional `clamp_to_road_band` for absolute plotting safety (disabled for intersection mode)

`audio/generation.py`

In `generate_multi_object_clip`:

changed default fallback `path_type` from 'straight' to 'bezier'

expanded `plot_kwargs` passed to plotting:

`intersection_benchmark`

`intersection_half_arm`

existing road/observer/lane params kept

`templates/index_batch.html`

Reworked benchmark UI structure:

added a third benchmark mode tab: Intersection benchmark (B9)

removed B9 from the regular multi-vehicle list (group-multi)

created dedicated group-intersection section with:

B9 checkbox

B9 intersection settings panel (extra-B9)

Updated benchmark mode switching logic:

switchBenchGroup now handles single, multi, and intersection

proper enable/disable/reset behavior across groups

auto-select behavior for B9 in intersection mode

Updated extra panel toggle behavior:

toggleBenchmarkExtra now resolves checked benchmark entries more robustly

Updated batch config payload:

reads selected benchMode

adds intersection-specific params under benchmarks.params:

intersection_benchmark

intersection_half_arm

existing is_crossing, lane width, etc. retained

If you want, I can also give you this same list as:

a compact “net final state only” changelog (without the intermediate iterations), or

a copy-paste commit-style changelog grouped by feature.